

Cross-Instrument Source Detection and Astrometry with a Joint Rubin–Euclid Foundation Model

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ABSTRACT

The Vera C. Rubin Observatory and *Euclid* survey overlapping sky at complementary wavelengths and resolutions, but their pixel-level data are rarely combined before catalog construction. We present JAISP, a joint Rubin–*Euclid* foundation model that ingests six Rubin optical bands (*ugrizy*) together with *Euclid* VIS and NISP (*YJH*) imagery at a common fused resolution, and learns a shared representation by masked reconstruction. We use the frozen representation to drive two downstream tasks on ~ 790 paired tiles in the ECDFS field: source detection with a center-prediction head, and astrometry via a multi-band joint centroid that fuses the sharp *Euclid* VIS localization across all instruments. The joint multi-band position reduces per-band centroid scatter relative to a single-band measurement, with the largest gains at low signal-to-noise ($\sim 2.4\text{--}2.5\times$ over a VIS-only anchor at $\text{SNR} = 5$ in a controlled source-injection test), and is validated against *Gaia* to $4.4\text{--}4.8\text{mas}$ at $G < 18.5$. We critically examine what the foundation contributes: through ablations and injection tests we find the cross-instrument gain is genuine but bounded by the VIS localization floor, and that detection completeness is suppressed near bright stars by the network’s signal-to-noise input normalization. Finally we report a first out-of-distribution test, applying the frozen model without retraining to the EDF-S field: detection and *Euclid* NISP astrometry transfer cleanly, [TODO: state final Rubin optical transfer result after real-variance rerun]. [TODO: one-sentence takeaway on the value of multi-instrument pixel-level fusion.]

Keywords: Astrometry — Surveys — Convolutional neural networks — Sky surveys — Astronomy data analysis

1. INTRODUCTION

Rubin and *Euclid* image overlapping sky with complementary strengths: Rubin provides deep six-band optical photometry at ground-based resolution, while *Euclid* VIS delivers diffraction-limited optical morphology and NISP adds near-infrared coverage. Jointly they constrain spectral energy distributions and positions better than either alone, yet standard pipelines process each instrument independently and combine only at the catalog level, discarding pixel-level cross-instrument information.

[TODO: Position relative to prior astronomical foundation / multimodal representation models (e.g. AstroCLIP, Multimodal Universe, AstroPT) and to classical multi-instrument forced photometry / forced astrometry. Make the novelty boundary explicit: pixel-level, multi-instrument, mixed-resolution fusion driving downstream detection and astrometry.]

In this paper we (i) describe JAISP, a joint Rubin–*Euclid* foundation model trained by masked reconstruction

on a common fused pixel grid (§3); (ii) use the frozen representation for source detection and for a multi-band joint-centroid astrometry that transfers the sharp *Euclid* VIS localization to bands where the source is faint (§4); (iii) validate positions against truth via source injection and against *Gaia*; and (iv) report a first out-of-distribution evaluation on a second field with no retraining (§5). Throughout we take a deliberately critical stance, using ablations and controlled injections to separate what the multi-instrument representation genuinely buys from what is already available in the single sharpest band.

Our main contributions are:

1. A multi-instrument, mixed-resolution pixel-level foundation model for Rubin+*Euclid*, with frozen-feature downstream heads.
2. A multi-band joint-centroid astrometry method and a controlled quantification, via source injection, of the cross-instrument localization gain and its VIS-imposed ceiling.

3. A first transfer (out-of-distribution) evaluation of the frozen stack on an independent field.
4. Honest characterization of failure modes: foundation-version insensitivity, the VIS localization floor, and bright-star detection suppression.

2. DATA

Our primary field is ECDFS (Rubin tract 5063), where we assemble ~ 790 paired Rubin+*Euclid* tiles. [TODO: exact tile count, sky area, depth, epoch.] For the out-of-distribution test (§5) we use 72 paired tiles in the EDF-S field (tract 2394).

Each tile carries six Rubin bands (*ugrizy*) and four *Euclid* bands (VIS, NISP *Y, J, H*). *Euclid* imagery is taken from the MER mosaics; VIS and NISP are sampled at $0.1'' \text{ px}^{-1}$ ($\sim 1084 \times 1084$ per tile), and Rubin at [TODO: Rubin pixel scale / tile shape ($6 \times 512 \times 512$)]. The native pixel scales and per-band World Coordinate Systems are retained; the foundation model resamples internally to a common fused scale of $0.4'' \text{ px}^{-1}$ (§3).

Per-pixel variance maps accompany the Rubin imagery and, in the primary field, the *Euclid* imagery. [TODO: Note: the EDF-S *Euclid* tiles used for the first OOD pass lacked variance maps; see §5. State final data state after real-variance tiles are obtained.]

For astrometric validation we use *Gaia* [TODO: DR] as an absolute reference, and [TODO: the *Euclid* MER catalog for detection completeness/purity]. We additionally construct a controlled source-injection set (§4) to measure positional error against known truth.

3. METHODS

3.1. The JAISP foundation model

JAISP encodes each band with a per-band convolutional stem, resamples the per-instrument streams to a common fused pixel scale of $0.4'' \text{ px}^{-1}$, and fuses them into a shared bottleneck representation. [TODO: Architecture detail: per-band stems, mixed-resolution stream depths from $\log_2(\text{scale ratio})$, ConvNeXt/transformer blocks, FiLM band conditioning, concat fusion (v9/v10).] The per-band stem normalizes its input by the per-pixel noise, $x = \text{image}/\text{rms}$, and clamps the result to a fixed signal-to-noise range before feature extraction; this choice is revisited in §4 as the origin of bright-star detection suppression.

The model is trained by masked reconstruction [TODO: (MAE objective, masking fraction, cross-instrument masking, Charbonnier+core-L2 loss for v10), epochs, optimizer, hardware]. All downstream heads operate on frozen features.

[TODO: Briefly: several architecture variants (v8 fine-scale, v9 concat fusion, v10 loss-shaped) were trained;

in-distribution probes and downstream metrics find them statistically equivalent (see §6). Report the production checkpoint used here (v10 warm-start).]

3.2. Detection head

We detect sources with a center-prediction (CenterNet-style) head on the frozen fused features, producing a normalized centroid and score per source. [TODO: Head architecture, self-training/pseudo-label procedure, operating point (conf threshold), NMS.] Detections are gated on the *Euclid* VIS signal and serve as seeds for astrometry.

3.3. Multi-band joint-centroid astrometry

Given a detection seed, we (i) refine the centroid classically on *Euclid* VIS, (ii) solve for a single *joint canonical* sky position by simultaneously fitting the source across all bands, weighting each band by its noise and accounting for its PSF, and (iii) measure each band’s classical centroid and record its offset from the joint canonical position. The joint canonical position is the cross-instrument estimate; the per-band offset is the astrometric residual we report.

Because the fit is dominated by the band with the sharpest, highest-SNR localization (typically *Euclid* VIS), the joint position transfers that localization to bands in which the source is faint or unresolved (§4). Per-source uncertainties follow from the χ^2 curvature of the joint fit and a per-band FWHM/(2.355 SNR) centroid-noise model.

We emphasize that step (i)–(iii) are classical given the detection seeds; the foundation’s role in astrometry is to supply the cross-instrument detections. A learned per-object position head was also trained and is discussed as an ablation in §6.

4. RESULTS ON ECDFS

4.1. Detection

On the ~ 790 ECDFS tiles the detector returns ~ 413 sources per tile at the chosen operating point. [TODO: Completeness/purity against a reference catalog; comparison to a classical VIS baseline (notebook 06); figure of the detection-vs-reference comparison.]

Detection completeness is reduced near bright stars: the source count per tile anticorrelates with the fraction of the tile under bright extended flux ($\text{corr} = -0.46$). We trace this to the per-band stem normalization (§3.1): a saturated star produces a wide region pinned at the clamped signal-to-noise ceiling, which dominates the group-wise feature normalization and depresses faint-source activations. We verify this is *not* a removable smooth background (subtracting a spatially varying

Figure 1. Astrometric residual versus signal-to-noise. [TODO: Final figure: per-band and pooled residual vs SNR; injection-truth overlay.]

background changes the count negligibly) and that it is intrinsic to the detector rather than a noise-model artifact. [TODO: Quantify against MER truth: are the lost sources real?]

4.2. Astrometry: per-band residual and SNR scaling

The joint-centroid astrometry reduces the pooled per-band median residual from ~ 50 mas (raw band-vs-canonical, dominated by per-band centroid scatter) to [TODO: state the corrected median, ~ 11 mas pooled, and the per-band table]. Residuals follow the expected $\text{FWHM}/(2.355 \text{ SNR})$ centroid-noise scaling (Fig. 1), reaching [TODO: floor at high SNR].

4.3. Source-injection truth test

To measure positional error against known truth we inject sub-pixel point sources across all ten bands at a range of signal-to-noise levels and recover their positions with the same pipeline. At $\text{SNR} = 5$ (Rubin) the classical single-band centroid errs by ~ 47 mas, the joint/canonical position by ~ 18 – 19 mas, approaching the VIS-only localization floor of ~ 17 mas; a position fitted jointly across all bands improves on a VIS-only anchor by ~ 2.4 – $2.5\times$ ($16.6 \rightarrow 6.7$ mas at peak $\text{SNR} = 5$). [TODO: Confirm numbers, add table, state caveats: idealized Gaussian PSFs, isolated sources, injected NISP sharper than the real MER PSF.] The test also bounds the gain: where a band is intrinsically sharper than VIS, transferring the VIS position degrades rather than improves it (§6).

4.4. Absolute validation against Gaia

Cross-matching the joint canonical positions to *Gaia* gives a bulk offset of $(+3.7, +1.9)$ mas and a per-star scatter of ~ 5.4 mas after bulk removal, improving to 4.4 – 4.8 mas for $G < 18.5$. [TODO: Gaia DR, matching radius, N, magnitude-binned table, spatial structure of the offset.]

5. OUT-OF-DISTRIBUTION TRANSFER: EDF-S

The value proposition of a foundation model is transfer without retraining. All results above are in the ECDFS field on which the downstream heads were tuned. Here we apply the frozen JAISP stack — foundation, detector, and joint-centroid astrometry — to 72 paired tiles in the independent EDF-S field (tract 2394) with no retraining or refitting.

[TODO: State the final data condition. The first pass used *Euclid* tiles without variance maps, requiring a

Figure 2. Per-band and SNR-binned astrometric residual, EDF-S (OOD) versus ECDFS. [TODO: final figure from notebook 15.]

median-absolute-deviation RMS fallback for VIS/NISP (Rubin variance was present and used directly). Results below are either (a) to be reported on real-variance tiles, or (b) reported with the fallback and the real-variance rerun as a robustness check.]

Detection transfers cleanly (456 sources per tile versus 413 on ECDFS; no processing errors), and *Euclid* NISP astrometry matches the in-distribution residual (35–41 mas versus 40 mas). The signal-to-noise scaling of the residual is reproduced. [TODO: Final Rubin optical (*griz*) transfer result: the first (fallback) pass showed a $\sim 1.5\times$ larger matched-SNR residual whose origin — the RMS fallback perturbing the joint-fit weighting versus genuine field differences — is resolved by the real-variance rerun. Report the resolved outcome here.]

The bright-star detection suppression (§4.1) reproduces and is somewhat stronger in EDF-S ($\text{corr} = -0.71$), consistent with its origin in the input normalization rather than the field. [TODO: MER-catalog completeness in EDF-S.]

6. DISCUSSION

6.1. What the multi-instrument representation actually buys

The cross-instrument gain is real but bounded. The joint position improves faint-band localization because it imports the sharpest available (*Euclid* VIS) constraint; it adds no precision beyond that band, and where another band is intrinsically sharper, transferring the VIS position is counterproductive (§4.3). The practical value is therefore largest exactly where it is most needed: measuring positions in bands where a source is too faint to localize on its own.

6.2. Foundation-version insensitivity

We trained several foundation variants (§3.1). In-distribution linear probes and downstream detection/astrometry metrics do not statistically distinguish them. [TODO: Quantify.] We argue the discriminating test for foundation quality is out-of-distribution transfer (§5) rather than in-distribution downstream accuracy, which saturates against label noise.

6.3. Label noise and the limits of in-distribution metrics

[TODO: Summarize the centroid-emulation finding: at the faint end the band-vs-canonical residual measures the model reproducing the canonical centroid definition, not positional truth; this is why injection and

Gaia (truth metrics) are necessary and in-distribution residuals are not sufficient.]

6.4. Bright-star detection suppression

The signal-to-noise input normalization that stabilizes training also lets a saturated star dominate the local feature statistics, suppressing faint detections nearby (§4.1). This is a completeness, not an astrometric-accuracy, effect. Mitigations — local background subtraction, a bright-star-aware rescue pass, or a normalization less sensitive to saturated outliers — are left to future work. [TODO: If MER shows the lost sources are real, quantify the completeness cost and any mitigation tested.]

6.5. Limitations

- Injection uses idealized PSFs and isolated sources; blends and realistic wings are not captured.
- Out-of-distribution evidence is from a single second field.
- [TODO: variance-map availability; Rubin optical transfer; others.]

7. SUMMARY

We presented JAISP, a joint Rubin–*Euclid* foundation model operating on a common fused pixel grid, and used its frozen representation for source detection and multi-band joint-centroid astrometry. Our main findings:

1. A joint multi-band position transfers the sharp *Euclid* VIS localization to faint bands, improv-

ing low-SNR positions by $\sim 2.4\text{--}2.5\times$ over a single-band measurement in controlled injection tests, and validating against *Gaia* to $4.4\text{--}4.8\text{ mas}$ at $G < 18.5$.

2. The gain is bounded by the VIS localization floor; the representation adds no precision beyond the sharpest contributing band.
3. The frozen stack transfers to an independent field without retraining for detection and *Euclid* NISP astrometry; [TODO: final Rubin optical transfer statement.]
4. Detection completeness is suppressed near bright stars by the input normalization, a characterized and mitigable effect.

[TODO: Closing sentence on the role of pixel-level multi-instrument fusion for Rubin+*Euclid* survey science, and next steps (additional fields, photometry, weak-lensing shapes).]

ACKNOWLEDGMENTS

[TODO: acknowledgments, facilities, software]

Facilities: Rubin Observatory, Euclid

Software: Astropy (Astropy Collaboration 2022), NumPy, SciPy, PyTorch, SEP (K. Barbary 2016)

REFERENCES

Astropy Collaboration. 2022, ApJ, 935, 167, doi: [10.3847/1538-4357/ac7c74](https://doi.org/10.3847/1538-4357/ac7c74)

Barbary, K. 2016, Journal of Open Source Software, 1, 58, doi: [10.21105/joss.00058](https://doi.org/10.21105/joss.00058)